Two-Dimensional Acceleration

Worked answer key: sketch, given, vector diagram, explained steps, final answer

Conventions used throughout

• **Notation:** $\vec{v}_1$ = initial velocity and $\vec{v}_2$ = final velocity. Question 9 writes the same things as $\vec{v}_i$ (i for initial) and $\vec{v}_f$ (f for final), so the key uses those symbols there. $\Delta\vec{v}$ = change in velocity, $\vec{a}_{av}$ = average acceleration, Δt = time taken. Bars around a vector, as in $|\Delta\vec{v}|$, mean its size only (no direction).

• **The big idea:** velocity has a direction, so an object that only changes direction (same speed) still has a change in velocity, and so it is accelerating.

• **Method (as the worksheet asks):** a subtraction of vectors is turned into an addition, $\vec{v}_2 - \vec{v}_1 = \vec{v}_2 + (-\vec{v}_1)$. Then the vectors are drawn tip to tail and the triangle is solved with trig: Pythagoras and tangent for a right angle, the cosine law and sine law otherwise.

• **Directions:** [N 11° E] means start facing north and turn 11° toward east. $-\vec{v}$ has the same size as $\vec{v}$ but points the opposite way. Diagrams are drawn to scale; full precision is kept until the final answer.

QUESTIONS 1–7

Average acceleration: turning bird

1. Write the equation that agrees with this definition: Acceleration is the rate of change of velocity.

A bird is flying at 3.4 m/s [S]. Over the next 1.5 s, it turns until it is flying at 3.4 m/s [W].

2. As a group, discuss: What is the direction of its average acceleration?

   a) west b) south c) south-west d) north-west e) nonsense, there is no acceleration because the speed is constant

3. Fill in the following quantities: $\vec{v}_1$ = _____ $\vec{v}_2$ = _____ Δt = _____

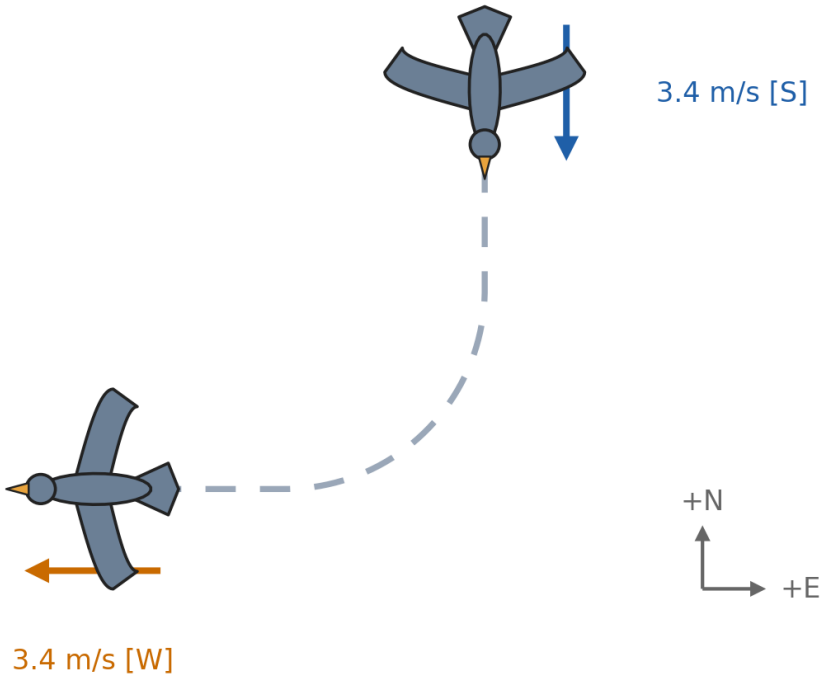
4. The numerator is vector subtraction – change it to vector addition (add the negative):

5. Now draw a vector diagram to do the vector addition in the numerator. Use trig (sine law or cosine law) to find the magnitude and direction of the resultant ($\Delta\vec{v}$).

6. Now complete the solution by dividing by the time.

7. Consider the direction of your answer. Explain the direction in terms of the physics involved.

Picture the situation



Seen from above: the bird is flying south, then over 1.5 s it curves around until it is flying west. Its speed stays 3.4 m/s the whole time; only its direction changes.

1. Definition as an equation

“Rate of change” means how much something changes divided by how long the change took. Here the thing that changes is velocity, so we divide the change in velocity by the time. Because velocity is a vector, the change is found by vector subtraction (final minus initial).

$$\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t} \qquad \text{and} \qquad \Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$

Put the second into the first: $\vec{a}_{av} = \dfrac{\vec{v}_2 - \vec{v}_1}{\Delta t}$

2. Is there an acceleration, and which way?

It is tempting to pick (e) because the speed doesn’t change. But velocity includes direction, and the direction changes from south to west, so the velocity changes. A change in velocity means there is an acceleration. Steps 5–7 below show the direction comes out north-west.

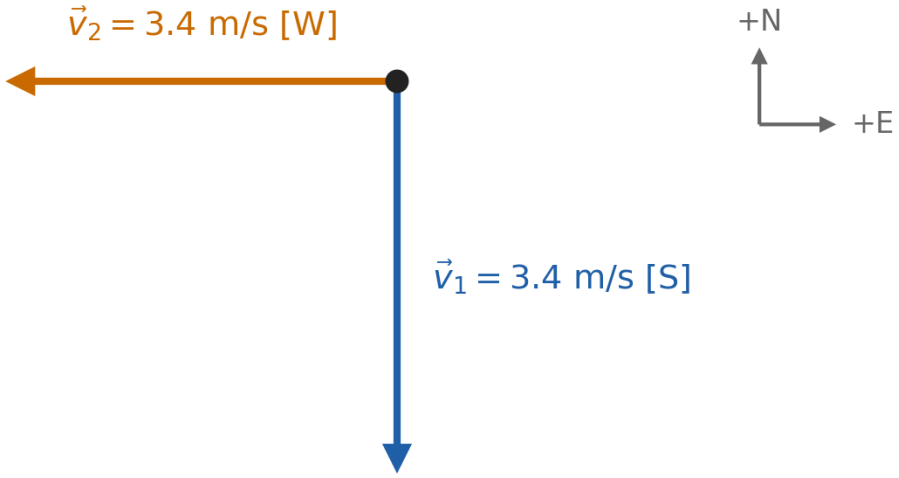
The answer is **(d) north-west**.

3. The quantities

Read them straight from the question. Velocities need a direction; time does not.

$$\vec{v}_1 = 3.4 \text{ m/s [S]} \qquad \vec{v}_2 = 3.4 \text{ m/s [W]} \qquad \Delta t = 1.5 \text{ s}$$

Diagram: the two velocities, tails together

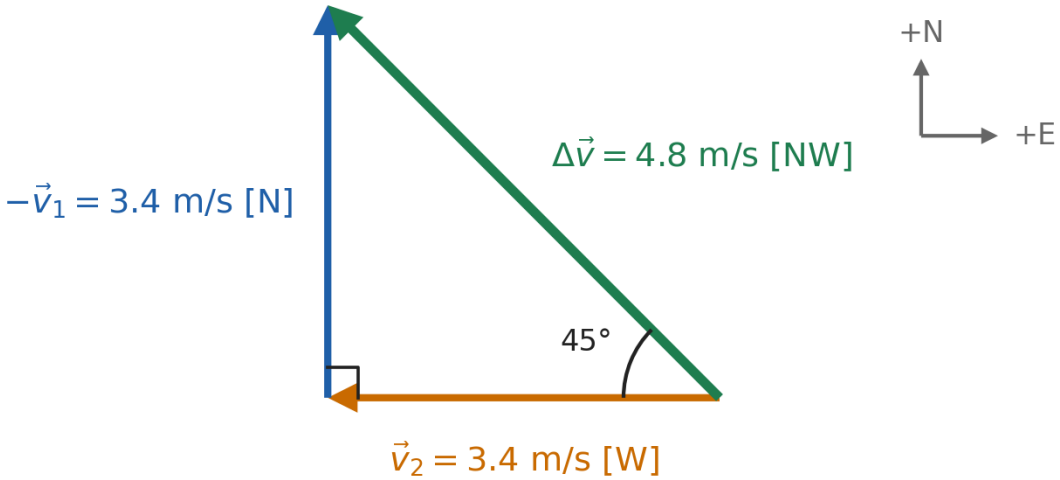


4. Turn the subtraction into an addition

We can't add arrows that are being subtracted, but subtracting a vector is the same as adding its opposite. The opposite of $\vec{v}_1 = 3.4$ m/s [S] is $-\vec{v}_1 = 3.4$ m/s [N] (same length, reversed).

$$\vec{a}_{av} = \frac{\vec{v}_2 + (-\vec{v}_1)}{\Delta t}$$

Diagram (Q5): $\vec{v}_2 + (-\vec{v}_1)$, tip to tail



Draw $\vec{v}_2$ first. Put the tail of $-\vec{v}_1$ on the tip of $\vec{v}_2$. The change in velocity, $\Delta\vec{v}$, goes from the very first tail to the very last tip.

5. Size and direction of $\Delta\vec{v}$

Now we solve the triangle. West and north are at 90° to each other, so it is a right triangle and the Pythagorean theorem gives the length of $\Delta\vec{v}$. For the direction, θ is the angle between $\Delta\vec{v}$ and west (marked 45° in the diagram). The opposite side is 3.4 (north) and the adjacent side is 3.4 (west), so θ = 45°: exactly halfway between west and north.

Formula (Pythagoras): $c^2 = a^2 + b^2 \Rightarrow c = \sqrt{a^2 + b^2}$

$$|\Delta\vec{v}| = \sqrt{(3.4)^2 + (3.4)^2} = \sqrt{11.56 + 11.56} = \sqrt{23.12} = 4.81 \text{ m/s}$$

Formula: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} \Rightarrow \theta = \tan^{-1}\left(\dfrac{\text{opposite}}{\text{adjacent}}\right)$

$$\theta = \tan^{-1}\left(\frac{3.4}{3.4}\right) = \tan^{-1}(1) = 45° \text{ north of west}$$

$$\Delta\vec{v} = 4.81 \text{ m/s [NW]}$$

6. Divide by the time

Finally divide the change in velocity by the 1.5 s it took. Dividing an arrow by a positive number makes it shorter but doesn't turn it, so the acceleration points the same way as $\Delta\vec{v}$.

Formula: $\vec{a}_{av} = \dfrac{\Delta\vec{v}}{\Delta t}$

$$\vec{a}_{av} = \frac{4.81 \text{ m/s [NW]}}{1.5 \text{ s}} = 3.21 \text{ m/s}^2 \text{ [NW]}$$

7. Why north-west makes sense

The acceleration has two jobs. First, it has to get rid of the bird's southward motion, and the only way to slow down motion to the south is to push **north**. Second, it has to build up motion to the west, which needs a push to the **west**. Doing both at once means pushing north-west. Notice also that north-west points to the inside of the turn, the way the bird is curving (look at the sketch).

Q1: $\vec{a}_{av} = \Delta\vec{v}/\Delta t = (\vec{v}_2 - \vec{v}_1)/\Delta t$

Q2: (d) north-west

Q3: $\vec{v}_1 = 3.4$ m/s [S], $\vec{v}_2 = 3.4$ m/s [W], $\Delta t = 1.5$ s

Q4: $\vec{a}_{av} = (\vec{v}_2 + (-\vec{v}_1))/\Delta t$

Q5: $\Delta\vec{v} = 4.8$ m/s [NW]

Q6: $\vec{a}_{av} = 3.2$ m/s^2 [NW]

Q7: north cancels the southward velocity; west builds the westward velocity

QUESTIONS 8–9

Turning car and finding initial velocity

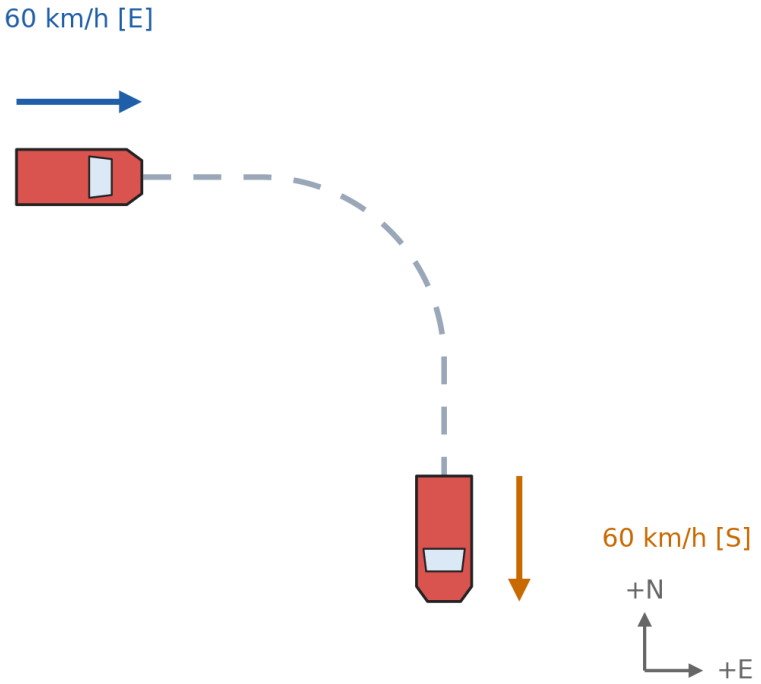
8. Say a car is travelling east at 60 km/h and executes a turn after which it is travelling 60 km/h [S]. In what direction is the acceleration?

9. You know that $\Delta\vec{v} = \vec{v}_f - \vec{v}_i$.

   a. Rewrite the equation to solve for initial velocity.

   b. If final velocity is 20 m/s [S] and the change in velocity is 10 m/s [NE], sketch a vector diagram (based on a.) that would allow you to find initial velocity.

Picture the situation



Question 8 seen from above: the car is heading east, then turns right until it is heading south, at the same speed.

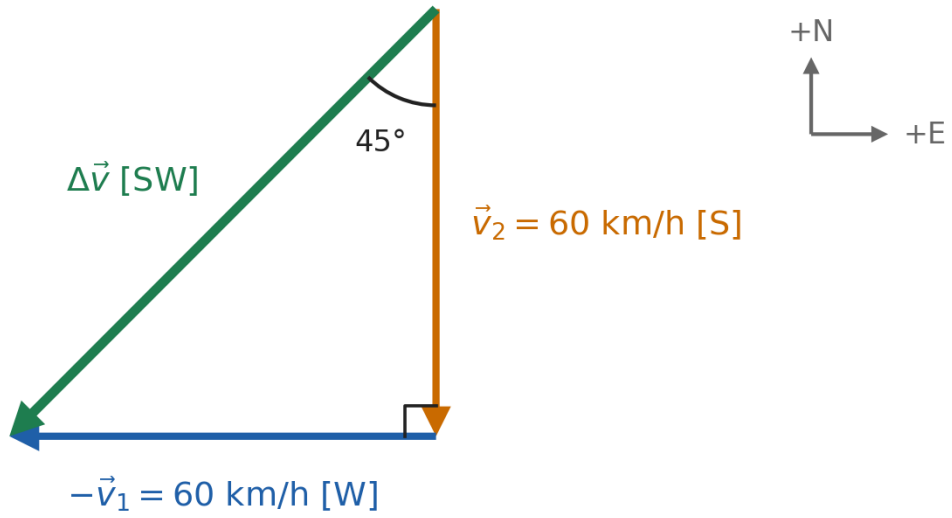
8. Direction of the car's acceleration

The acceleration points the same way as the change in velocity, $\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$. Here $\vec{v}_2$ = 60 km/h [S], and reversing $\vec{v}_1$ = 60 km/h [E] gives $-\vec{v}_1$ = 60 km/h [W]. Adding a south arrow and a west arrow of equal length gives an arrow exactly between them.

$$\Delta\vec{v} = 60 \text{ km/h [S]} + 60 \text{ km/h [W]} \Rightarrow \text{direction [SW]}$$

No time is given, so we can only find the direction, not the size of the acceleration. As with the bird, SW points to the inside of the turn.

Diagram (Q8): $\vec{v}_2 + (-\vec{v}_1)$, tip to tail



9(a) Solve for the initial velocity

Treat it like ordinary algebra: add $\vec{v}_i$ to both sides so it is no longer negative, then subtract $\Delta\vec{v}$ from both sides. To draw it, write the subtraction as adding the opposite.

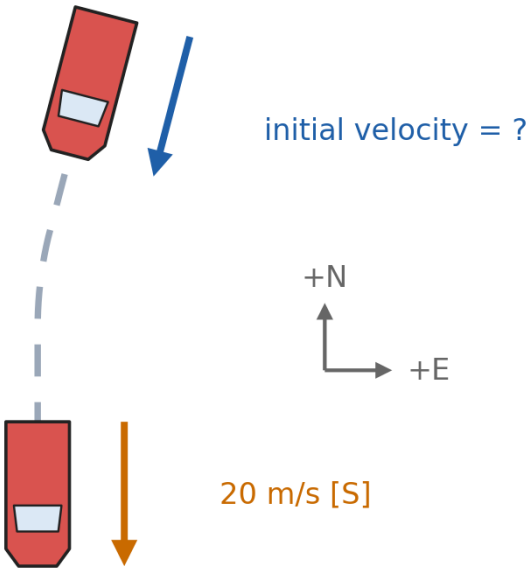
Formula: $\Delta\vec{v} = \vec{v}_f - \vec{v}_i$

Add $\vec{v}_i$ to both sides: $\Delta\vec{v} + \vec{v}_i = \vec{v}_f$

Subtract $\Delta\vec{v}$ from both sides: $\vec{v}_i = \vec{v}_f - \Delta\vec{v}$

Write the subtraction as adding the opposite: $\vec{v}_i = \vec{v}_f + (-\Delta\vec{v})$

Picture the situation

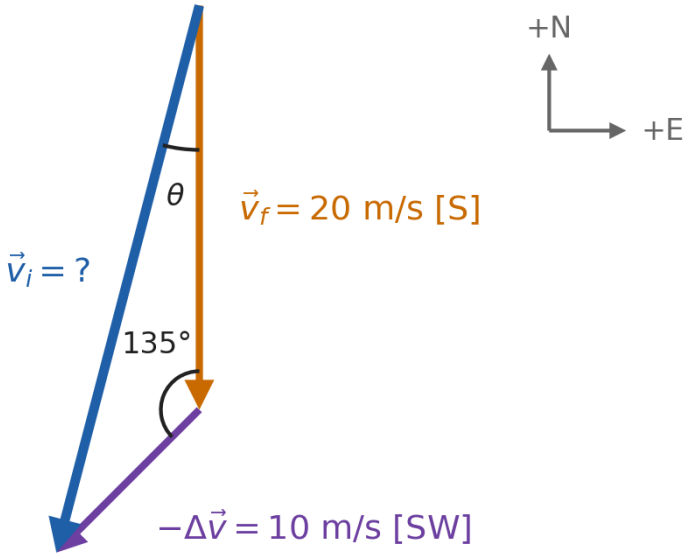


Question 9: an object ends up moving 20 m/s [S]. We know how its velocity changed (10 m/s [NE]) and want to work backward to how it was moving before.

9(b) The vector diagram

$-\Delta\vec{v}$ is 10 m/s [SW] (the opposite of NE). Draw $\vec{v}_f$ = 20 m/s [S], put the tail of $-\Delta\vec{v}$ on its tip, and $\vec{v}_i$ goes from the first tail to the last tip. At the tip of $\vec{v}_f$, looking back along $\vec{v}_f$ points north; $-\Delta\vec{v}$ points 45° away from south, so the inside angle is 180° − 45° = 135°.

Diagram (Q9b): $\vec{v}_f + (-\Delta\vec{v})$, tip to tail



9(b) Optional: solving the triangle

The question only asks for the sketch, but here is how to finish it. This triangle has no right angle, so Pythagoras doesn't work. We know two sides (20 and 10) and the angle between them (135°), which is exactly the situation for the cosine law. In the cosine law, C is the angle we know (135°), c is the side across from it ($\vec{v}_i$, unknown), and a and b are the other two sides (20 and 10). Note that cos 135° is negative (−0.7071), so "minus 2ab cos C" turns into a plus. Then the sine law gives the angle θ at the start: it matches each angle with the side across from it (θ is across from the 10 side; 135° is across from the 27.98 side).

Formula (cosine law): $c^2 = a^2 + b^2 - 2ab\cos C$

$|\vec{v}_i|^2 = 20^2 + 10^2 - 2(20)(10)\cos 135°$

$|\vec{v}_i|^2 = 400 + 100 - (400)(-0.7071) = 500 + 282.84 = 782.84$

Square root both sides: $|\vec{v}_i| = \sqrt{782.84} = 27.98$ m/s

Formula (sine law): $\frac{\sin A}{a} = \frac{\sin C}{c}$

$\frac{\sin\theta}{10} = \frac{\sin 135°}{27.98}$

Multiply both sides by 10: $\sin\theta = \frac{10\sin 135°}{27.98} = \frac{7.071}{27.98} = 0.2527$

Inverse sine: $\theta = \sin^{-1}(0.2527) = 14.64°$ west of south

Q8: [SW]

Q9(a): $\vec{v}_i = \vec{v}_f - \Delta\vec{v} = \vec{v}_f + (-\Delta\vec{v})$

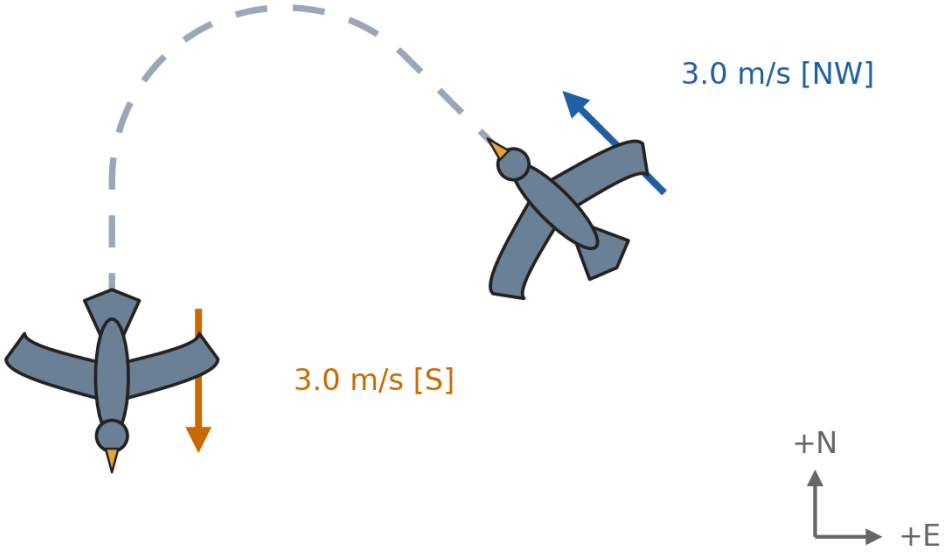
Q9(b): the diagram above; solving it gives $\vec{v}_i$ = 28 m/s [S 15° W]

PRACTICE 1

Bird turns from NW to S

A bird travelling 3.0 m/s NW turns and heads south at the same speed. Determine its change in velocity.

Picture the situation

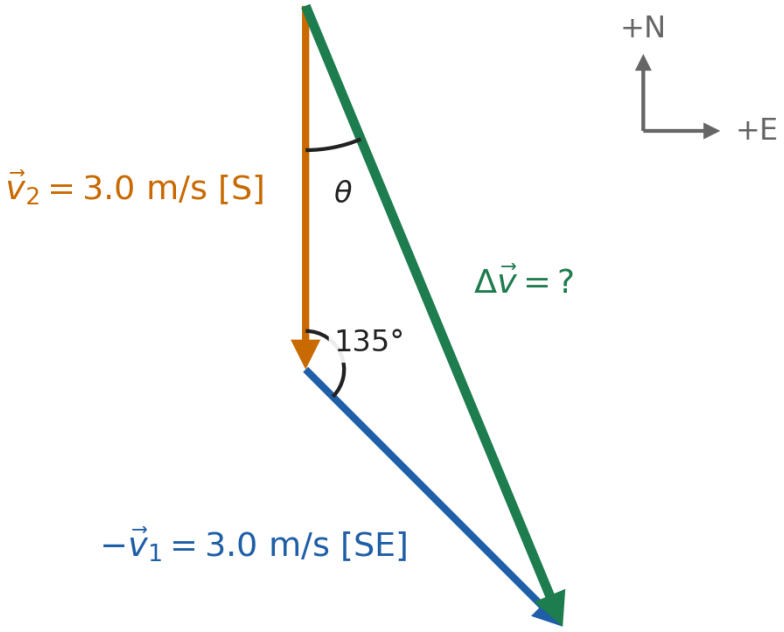


Seen from above: the bird starts flying north-west, swings around, and ends up flying south at the same speed.

GIVEN

$\vec{v}_1 = 3.0$ m/s [NW] $\quad \vec{v}_2 = 3.0$ m/s [S] $\quad \Delta\vec{v} = ?$

Diagram: $\vec{v}_2 + (-\vec{v}_1)$, tip to tail



Step 1 – Set up the addition and find the angle in the triangle

Write $\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$. Reversing $\vec{v}_1 = 3.0$ m/s [NW] gives $-\vec{v}_1 = 3.0$ m/s [SE]. At the tip of $\vec{v}_2$, looking back along $\vec{v}_2$ points north; $-\vec{v}_1$ points 45° away from south, so the inside angle is $180° - 45° = 135°$.

There is no right angle, so we use the cosine law (two sides and the angle between them).

Step 2 – Size of $\Delta\vec{v}$ with the cosine law

In the cosine law, C is the angle we know (135°), c is the side across from it ($\Delta\vec{v}$, what we want), and a and b are the other two sides (3.0 and 3.0). cos 135° is negative (−0.7071), so the last term turns into a plus.

Formula (cosine law): $c^2 = a^2 + b^2 - 2ab\cos C$

$$|\Delta\vec{v}|^2 = (3.0)^2 + (3.0)^2 - 2(3.0)(3.0)\cos 135°$$

$$|\Delta\vec{v}|^2 = 9 + 9 - (18)(-0.7071) = 18 + 12.73 = 30.73$$

Square root both sides: $|\Delta\vec{v}| = \sqrt{30.73} = 5.54$ m/s

The change is bigger than either velocity, because the bird almost reverses direction.

Step 3 – Direction with the sine law

θ is the angle at the start of the triangle, between $\vec{v}_2$ (south) and $\Delta\vec{v}$. The side opposite θ is $-\vec{v}_1$ (3.0), and the side opposite the 135° angle is $\Delta\vec{v}$ (5.543). The sine law matches each side with its opposite angle:

Formula (sine law): $\frac{\sin A}{a} = \frac{\sin C}{c}$

$$\frac{\sin\theta}{3.0} = \frac{\sin 135°}{5.543}$$

Multiply both sides by 3.0:

$$\sin\theta = \frac{3.0\sin 135°}{5.543} = \frac{2.121}{5.543} = 0.3827$$

Inverse sine: $\theta = \sin^{-1}(0.3827) = 22.5°$ east of south

Quick check: the triangle has two equal sides (3.0 and 3.0), so its other two angles are equal: (180° − 135°)/2 = 22.5°. Same answer.

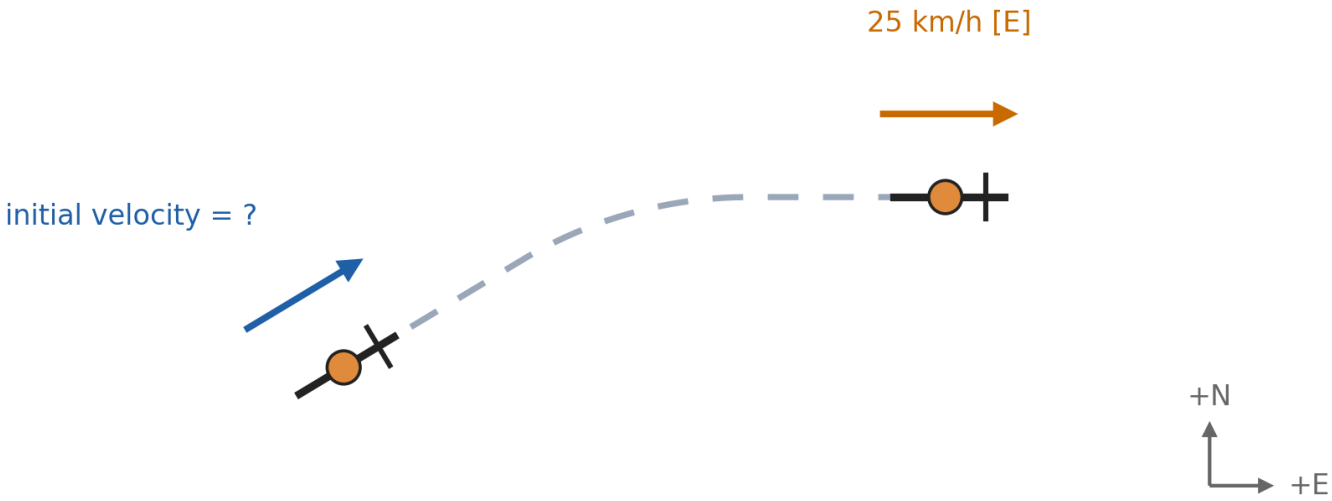
$\Delta\vec{v}$ = 5.5 m/s [S 23° E] (the angle is exactly 22.5°)

PRACTICE 2

Cyclist: find the initial velocity

After completing a turn, a cyclist is travelling east at 25 km/h. If her change in velocity was 15 km/h south, determine her initial velocity.

Picture the situation



Seen from above: the cyclist finishes a turn heading east. We work backward to find which way, and how fast, she was going before the turn.

GIVEN

$\vec{v}_2 = 25$ km/h [E] $\quad \Delta\vec{v} = 15$ km/h [S] $\quad \vec{v}_1 = ?$

Step 1 – Rearrange for the initial velocity

Solve for $\vec{v}_1$ the same way as Q9(a). Subtracting $\Delta\vec{v}$ = 15 km/h [S] is the same as adding 15 km/h [N].

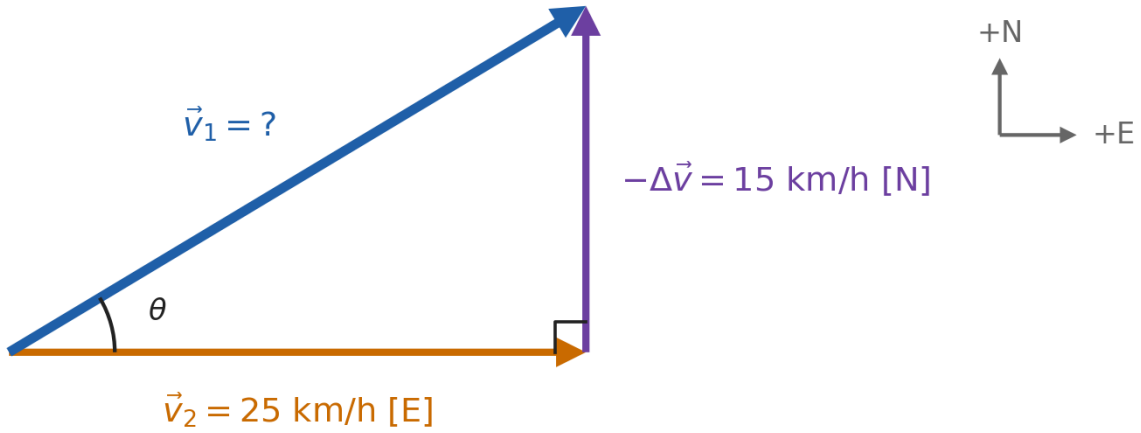
Formula: $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$

Add $\vec{v}_1$ to both sides: $\Delta\vec{v} + \vec{v}_1 = \vec{v}_2$

Subtract $\Delta\vec{v}$ from both sides: $\vec{v}_1 = \vec{v}_2 - \Delta\vec{v}$

$\vec{v}_1 = \vec{v}_2 + (-\Delta\vec{v}) = 25$ km/h [E] $+ 15$ km/h [N]

Diagram: $\vec{v}_2 + (-\Delta\vec{v})$, tip to tail



Step 2 – Solve the right triangle

East and north are perpendicular, so this is a right triangle. Pythagoras gives the size. For the angle measured from east, the opposite side is 15 (north) and the adjacent side is 25 (east), so use tangent.

Formula (Pythagoras): $c^2 = a^2 + b^2 \Rightarrow c = \sqrt{a^2 + b^2}$

$$|\vec{v}_1| = \sqrt{25^2 + 15^2} = \sqrt{625 + 225} = \sqrt{850} = 29.15 \text{ km/h}$$

Formula: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} \Rightarrow \theta = \tan^{-1}\left(\dfrac{\text{opposite}}{\text{adjacent}}\right)$

$$\theta = \tan^{-1}\left(\frac{15}{25}\right) = \tan^{-1}(0.6) = 30.96° \text{ north of east}$$

$\vec{v}_1$ = 29 km/h [E 31° N]

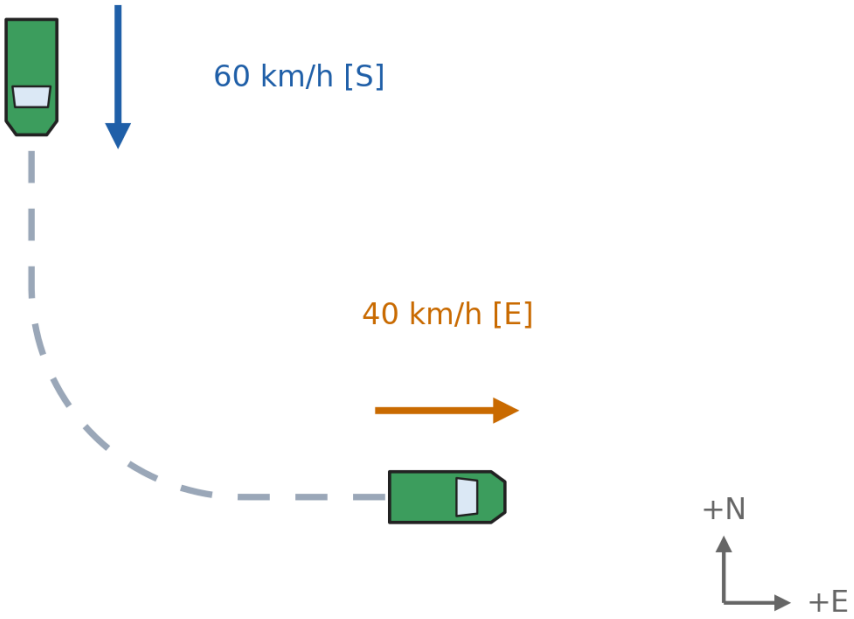
Check: before the turn she was going partly north. The southward change cancelled that northward part and left her heading due east.

PRACTICE 3

Go-cart: average acceleration

A go cart rounding a curve goes from an initial velocity of 60 km/h [S] to 40 km/h [E]. If the curve takes 2.0 s, determine the cart's acceleration.

Picture the situation



Seen from above: the go-cart comes down the track heading south, rounds the curve, and leaves heading east, slower than before.

GIVEN

$\vec{v}_1 = 60$ km/h [S] $\quad \vec{v}_2 = 40$ km/h [E] $\quad \Delta t = 2.0$ s

Step 1 – Change in velocity

$\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$. Reversing 60 km/h [S] gives $-\vec{v}_1$ = 60 km/h [N]. East and north are perpendicular, so the triangle has a right angle: Pythagoras for the size, tangent for the angle measured from east (opposite 60, adjacent 40).

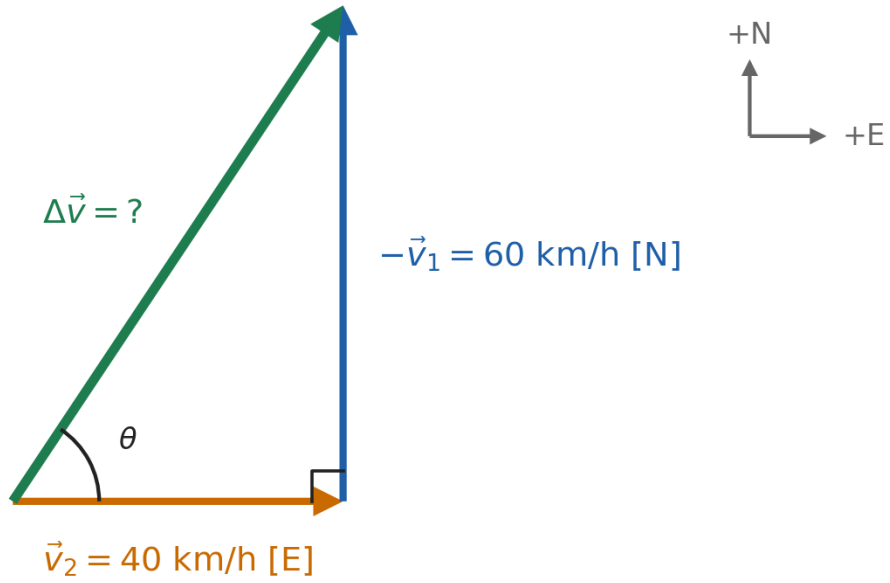
Formula (Pythagoras): $c^2 = a^2 + b^2 \Rightarrow c = \sqrt{a^2 + b^2}$

$$|\Delta\vec{v}| = \sqrt{40^2 + 60^2} = \sqrt{1600 + 3600} = \sqrt{5200} = 72.11 \text{ km/h}$$

Formula: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} \Rightarrow \theta = \tan^{-1}\left(\dfrac{\text{opposite}}{\text{adjacent}}\right)$

$$\theta = \tan^{-1}\left(\frac{60}{40}\right) = \tan^{-1}(1.5) = 56.31° \text{ north of east}$$

Diagram: $\vec{v}_2 + (-\vec{v}_1)$, tip to tail



Step 2 – Divide by the time

Acceleration is the change in velocity divided by the time, and the direction stays the same. The answer comes out in (km/h)/s, which means “km/h of velocity change every second”. To convert to the usual m/s², divide by 3.6, because 1 km/h = 1000 m ÷ 3600 s = 1/3.6 m/s.

Formula: $\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t}$

$$\vec{a}_{av} = \frac{72.11 \text{ km/h}}{2.0 \text{ s}} = 36.06 \text{ (km/h)/s [E 56° N]}$$

$$36.06 \text{ (km/h)/s} \div 3.6 = 10.02 \text{ m/s}^2$$

$\vec{a}_{av}$ = 36 (km/h)/s [E 56° N], which is 1.0×10^1 m/s^2 [E 56° N]

PRACTICE 4

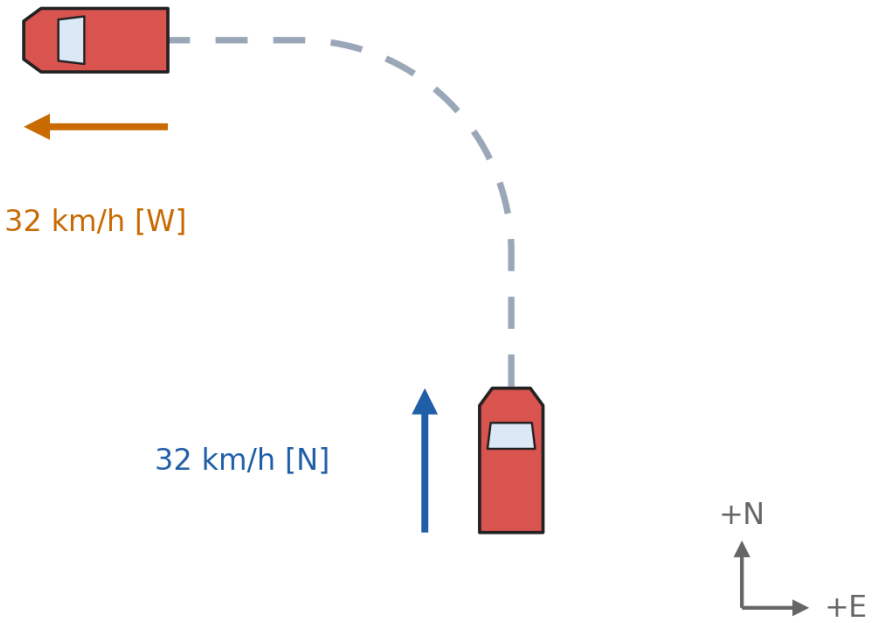
Car: change in velocity and acceleration

A car, moving initially at 32 km/h [N], turns a corner and continues at 32 km/h [W]. The turn takes 3.0 s to complete. Find

a. the change in velocity

b. the average acceleration during the turn.

Picture the situation

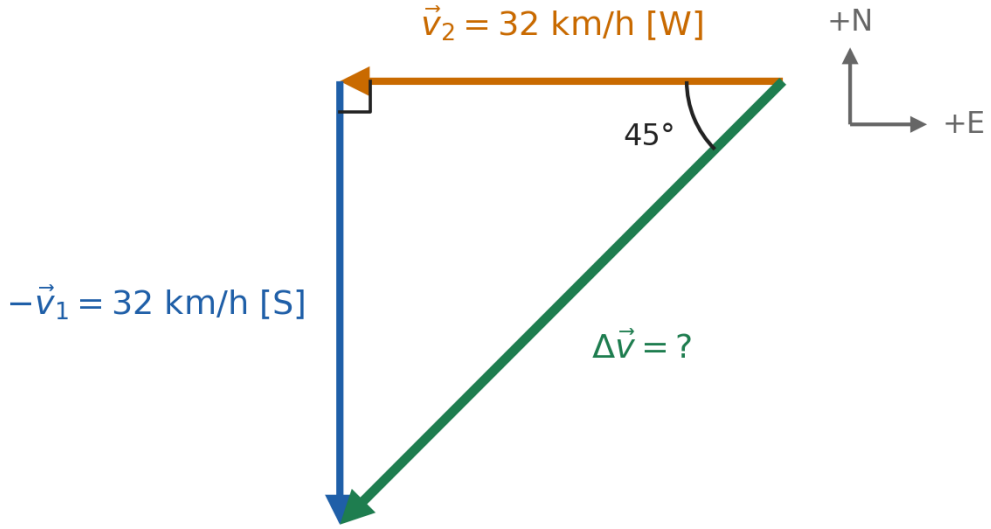


Seen from above: the car drives north, turns left at the corner, and continues west at the same speed.

GIVEN

$\vec{v}_1 = 32$ km/h [N] $\vec{v}_2 = 32$ km/h [W] $\Delta t = 3.0$ s

Diagram: $\vec{v}_2 + (-\vec{v}_1)$, tip to tail



(a) Change in velocity

$\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$, with $-\vec{v}_1 = 32$ km/h [S]. West and south are perpendicular, so Pythagoras gives the size. θ is the angle between $\Delta\vec{v}$ and west (marked 45° in the diagram). The opposite side is 32 (south) and the adjacent side is 32 (west), so θ = 45°: the direction is exactly south-west.

Formula (Pythagoras): $c^2 = a^2 + b^2 \Rightarrow c = \sqrt{a^2 + b^2}$

$$|\Delta\vec{v}| = \sqrt{32^2 + 32^2} = \sqrt{1024 + 1024} = \sqrt{2048} = 45.25 \text{ km/h}$$

Formula: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} \Rightarrow \theta = \tan^{-1}\left(\dfrac{\text{opposite}}{\text{adjacent}}\right)$

$$\theta = \tan^{-1}\left(\frac{32}{32}\right) = \tan^{-1}(1) = 45° \text{ south of west}$$

(b) Average acceleration

Divide the change in velocity by the 3.0 s; dividing by a positive time keeps the direction. Then convert to m/s² by dividing by 3.6.

Formula: $\vec{a}_{av} = \dfrac{\Delta \vec{v}}{\Delta t}$

$$\vec{a}_{av} = \frac{45.25 \text{ km/h [SW]}}{3.0 \text{ s}} = 15.08 \text{ (km/h)/s [SW]}$$

$$15.08 \div 3.6 = 4.19 \text{ m/s}^2$$

SW points to the inside of the left turn, as expected.

(a) $\Delta \vec{v}$ = 45 km/h [SW]

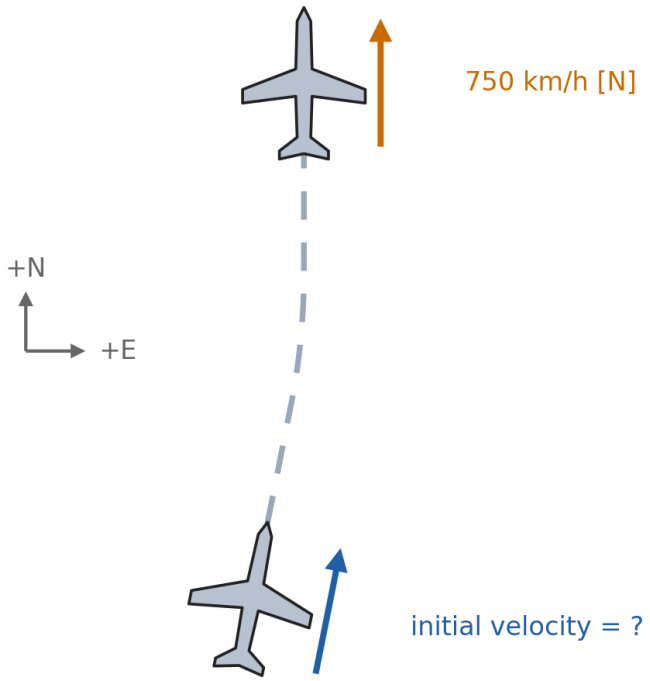
(b) $\vec{a}_{av}$ = 15 (km/h)/s [SW], which is 4.2 m/s^2 [SW]

PRACTICE 5

Fighter jet: initial velocity

A fighter jet comes out of a curve traveling 750 km/h [N]. During the curve, it experienced an acceleration of 3 g's (three times acceleration due to gravity) [SW] for a duration of 2.5 s. What was the jet's initial velocity?

Picture the situation



Seen from above: the jet comes out of a curve heading north. We know its acceleration during the curve and work backward to how it was moving when the curve began.

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$\vec{v}_2 = 750$ km/h [N] $\quad \vec{a}_{av} = 3g$ [SW] $\quad \Delta t = 2.5$ s

$$\vec{v}_2 = 750 \div 3.6 = 208.33 \text{ m/s [N]}$$

$$|\vec{a}_{av}| = 3(9.81 \text{ m/s}^2) = 29.43 \text{ m/s}^2$$

Step 1 – How much the velocity changed

The units must match before we combine anything, so the speed is converted to m/s above. Then rearrange $\vec{a}_{av} = \Delta\vec{v}/\Delta t$ to get $\Delta\vec{v} = \vec{a}_{av}\Delta t$. Multiplying by a positive time keeps the direction, so $\Delta\vec{v}$ points SW like $\vec{a}_{av}$.

Formula: $\vec{a}_{av} = \dfrac{\Delta\vec{v}}{\Delta t}$

Multiply both sides by Δt: $\Delta\vec{v} = \vec{a}_{av}\,\Delta t$

$\Delta\vec{v} = (29.43\ \text{m/s}^2)(2.5\ \text{s}) = 73.58\ \text{m/s [SW]}$

Step 2 – Set up the initial velocity

As in Q9, $\vec{v}_1 = \vec{v}_2 + (-\Delta\vec{v})$, and $-\Delta\vec{v} = 73.58$ m/s [NE]. Draw $\vec{v}_2$ north, then $-\Delta\vec{v}$ pointing NE from its tip. At the tip of $\vec{v}_2$, looking back along $\vec{v}_2$ points south; $-\Delta\vec{v}$ points 45° away from north, so the inside angle is 180° − 45° = 135°. There is no right angle, so use the cosine law: C = 135°, a and b are the two known sides (208.33 and 73.58), and c is the side across from 135° ($\vec{v}_1$). cos 135° is negative, so the last term becomes a plus. Then the sine law gives θ (east of north), which is across from the 73.58 side.

Formula (cosine law): $c^2 = a^2 + b^2 - 2ab\cos C$

$$|\vec{v}_1|^2 = (208.33)^2 + (73.58)^2 - 2(208.33)(73.58)\cos 135°$$

$$|\vec{v}_1|^2 = 43\,403 + 5413 - (30\,656)(-0.7071)$$

$$|\vec{v}_1|^2 = 48\,816 + 21\,677 = 70\,493$$

Square root both sides: $|\vec{v}_1| = \sqrt{70\,493} = 265.5$ m/s

In km/h: $265.5 \times 3.6 = 955.8$ km/h

Formula (sine law): $\dfrac{\sin A}{a} = \dfrac{\sin C}{c}$

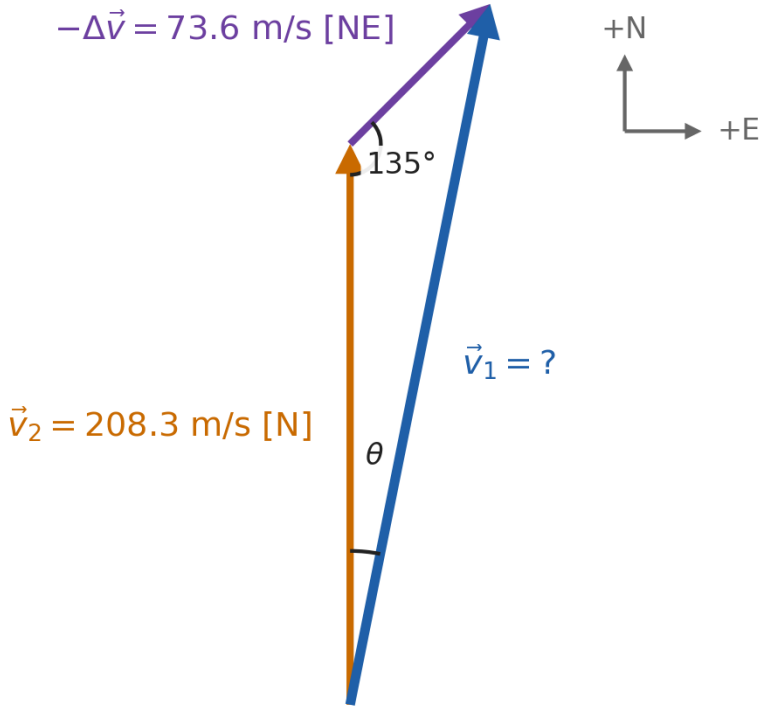
$$\frac{\sin\theta}{73.58} = \frac{\sin 135°}{265.5}$$

Multiply both sides by 73.58:

$$\sin\theta = \frac{73.58\sin 135°}{265.5} = \frac{52.03}{265.5} = 0.1959$$

Inverse sine: $\theta = \sin^{-1}(0.1959) = 11.30°$ east of north

Diagram: $\vec{v}_2 + (-\Delta\vec{v})$, tip to tail



$\vec{v}_1 = 2.7 \times 10^2$ m/s [N 11° E] (about 9.6×10^2 km/h)

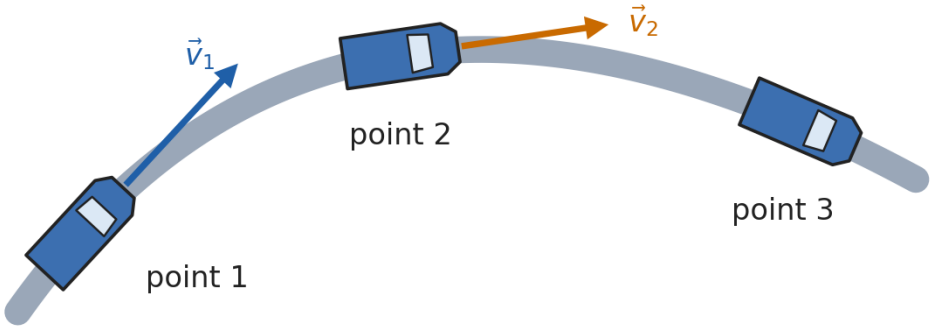
Check: the jet was pushed toward the south-west during the curve. So before the curve it must have been going faster than at the end (266 m/s vs 208 m/s) and slightly more toward the east. That matches.

PRACTICE 6

Racing car: extend the velocity pattern

The diagram below a racing car's position at three equally spaced points in time. It also shows the car's instantaneous velocity vectors at points 1 and 2. If the acceleration of the car is uniform, determine, using an accurate vector scale diagram, the instantaneous velocity of the car at point 3. Scale: 1 cm = 5.0 m/s.

Picture the situation



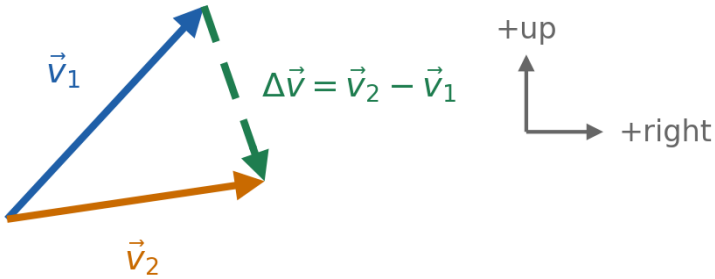
The car at three moments, equally spaced in time, as it goes around a curve. The arrows used below are $\vec{v}_1 \approx 13$ m/s right and 14 m/s up, and $\vec{v}_2 \approx 17$ m/s right and 2.5 m/s up.

Step 1 – The key physics

"Uniform acceleration" means the velocity changes by the same amount, in the same direction, in every equal time interval. The time from point 1 to 2 equals the time from point 2 to 3, so whatever change turned $\vec{v}_1$ into $\vec{v}_2$ must happen again to turn $\vec{v}_2$ into $\vec{v}_3$.

$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1 \qquad \vec{v}_3 = \vec{v}_2 + \Delta\vec{v}$$

Diagram 1: find $\Delta\vec{v}$ (tails together)



Copy $\vec{v}_1$ and $\vec{v}_2$ with their tails together. $\Delta\vec{v}$ is the arrow from the tip of $\vec{v}_1$ to the tip of $\vec{v}_2$ (because $\vec{v}_1 + \Delta\vec{v} = \vec{v}_2$). $\vec{v}_1 \approx 19.1$ m/s (3.8 cm), $\vec{v}_2 \approx 17.2$ m/s (3.4 cm).

Diagram 2: add the same Δv⃗ to v⃗₂



Step 2 – Build and measure v⃗₃

Copy Δv⃗ (same length, same direction) onto the tip of $\vec{v}_2$. The arrow from the tail of $\vec{v}_2$ to the new tip is $\vec{v}_3$. From Diagram 1, Δv⃗ is 17 − 13 = 4 m/s right and 2.5 − 14 = −11.5 m/s, so 11.5 m/s down. Adding it to $\vec{v}_2$: 17 + 4 = 21 m/s right and 2.5 − 11.5 = −9 m/s, so 9 m/s down. On your scale diagram $\vec{v}_3$ should measure about 4.6 cm at about 23° below the rightward direction. The calculation below checks the ruler and protractor reading:

Formula (Pythagoras): $c^2 = a^2 + b^2 \Rightarrow c = \sqrt{a^2 + b^2}$

$|\vec{v}_3| \approx \sqrt{21^2 + 9^2} = \sqrt{441 + 81} = \sqrt{522} = 22.8 \text{ m/s}$

Scale length: $22.8 \div 5.0 = 4.6 \text{ cm}$

Formula: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}} \Rightarrow \theta = \tan^{-1}\left(\dfrac{\text{opposite}}{\text{adjacent}}\right)$

$\theta \approx \tan^{-1}\left(\dfrac{9}{21}\right) = \tan^{-1}(0.429) = 23°$

$\vec{v}_3 = \vec{v}_2 + \Delta\vec{v} \approx$ 23 m/s, about 23° below the rightward direction